Prompt 2 Genetic Algorithm Optimization

Prompt 2: Genetic Algorithm Optimization

Step 1: Check Initial Results

Read the text file containing specs & the initial Design Point provided to you .

- **If you find all the specs achieved:** Output exactly `"All specs achieved, nothing else to do"` in the chat and stop execution.
- **If you find that some of the specs weren't achieved:** Proceed immediately to Step 2.

Step 2: Optimization Rules & Environment

If optimization is required, you must strictly follow these rules:

1. **Do not modify or touch** any of the existing files in the project folder.
2. Use the Python environment called `Thesis_optimizer_Sandbox_Copy`.
3. You are allowed to create new Python files (`.py`) and Jupyter Notebooks (`.ipynb`), as long as they do not modify or affect any of the existing files in the project folder.
4. You are allowed to create notepads and text files, as long as they do not modify or affect any of the existing files in the project folder.

Step 3: Optimizer Setup

You will be optimizing the circuit using the Genetic Algorithm Optimizer located in the project folder.

You will use the `UniversalGeneticOptimizer` tool from `GA_Optimizer.py`:

```
from GA_Optimizer import UniversalGeneticOptimizer
```

You will also have access to the simulator calling functions (Circuit Wrappers) :

```python
from LTspice_Circuit_Wrappers import evaluate_5t_ota_ltspice
from SpiceFunctions import SpiceEvaluator, SpiceWaveformReader

LTSPICE_PATH =
r"C:\Users\Mohamed\AppData\Local\Programs\ADI\LTspice\LTspice.exe"
BASE_DIR =
r"C:\Users\Mohamed\Desktop\ML\CAD_projects\Netlists\GF_180nm_tech_work\pr
oject"
```

Step 4: Initial Point Extraction

Read the .txt file given to you again and extract the Design Initial Design Point parameters .

Read the Specs.md file as well and extract the Target Specs:

```
Initial Design Point Transistor Dimensions:
W_d  =  # Read this number and store it in python variable Wd
L_d  =   # Read this number and store it in python variable Ld
W_m  = # Read this number and store it in python variable Wm
L_m  =  # Read this number and store it in python variable Lm
W_ss = # Read this number and store it in python variable Wss
L_ss =  # Read this number and store it in python variable Lss
W_B  =  # Read this number and store it in python variable WB



--- SPECIFICATION PERFORMANCE ---
  Spec            | Target       | Achieved     | Status
  ----------------------------------------------------------
  Av              |              |              |  [PASS/FAIL]
  UGF             |              |              |  [PASS/FAIL]
  CMRR            |              |              |  [PASS/FAIL]
  PM              |              |              |  [PASS/FAIL]
  Pw              |              |              |  [PASS/FAIL]
  Av_VICM_MIN     |              |              |  [PASS/FAIL]
  Av_VICM_MAX     |              |              |  [PASS/FAIL]
=================================================================
```

Store the Target specs and their values in Python variables as well .

Like this :

```
# ═══════════════════════════════════════════════════════════════════════
#  Store the needed Variables
# ═══════════════════════════════════════════════════════════════════════
#### SPECS
Av_TG=  # Read from the txt file  specification performance table  target column or from spec file
UGF_TG= # Read from the txt file  specification performance table  target column or from spec file
CMRR_TG=# Read from the txt file  specification performance table  target column or from spec file
Vicmmax_TG=# Read from the txt file  specification performance table target column or from spec file
Vicmmin_TG=# Read from the txt file  specification performance table target column or from spec file
PM_TG=# Read from the txt file  specification performance table  target column or from spec file
Pw_TG=# Read from the txt file  specification performance table  target column or from spec file
Av_VICM_MIN=# Read from the txt file  specification performance table target column or from spec file
Av_VICM_MAX= #Read from the txt file  specification performance table target column or from spec file
TGT_VICM_MIN=Vicmmin_TG
TGT_VICM_MAX=Vicmmax_TG
TGT_VOUT_MIN=0.5 # Use this number just to run the GA+Simulator loop , they are not mentioned
TGT_VOUT_MAX=2.5 # Use this number just to run the GA+Simulator loop , they are not mentioned


CL= # use the capacitance value that is given in the spec file
VDD=  # use the VDD value that is given in the spec file
IB= # use the current value that is given in the spec file
```

```
#### FEATURE VARIABLES
Ld=  # Taken from Initial Design point .txt file
Wd= # Taken from Initial Design point .txt file
Lm= # Taken from Initial Design point .txt file
Wm= # Taken from Initial Design point .txt file
Lss= # Taken from Initial Design point .txt file
Wss=# Taken from Initial Design point .txt file
WB = # Taken from Initial Design point .txt file
Ib= IB
Vicm= # Taken from specs.md file (nominal value for Vicm)
```

Step 5: Define your mathematical problem

Define your targets vector

```
# ══════════════════════════════════════════════════════════════════════
#  Define Target Vectors
# ══════════════════════════════════════════════════════════════════════
#### SPECS
y_spec = {
    'Av': Av_TG,
    'UGF': UGF_TG,
    'CMRR': CMRR_TG,
    'PM': PM_TG,
    'Pw': Pw_TG,
    'Av_VICM_MIN': Av_VICM_MIN,
    'Av_VICM_MAX': Av_VICM_MAX,
}

maximize_metrics = ['Av', 'UGF', 'CMRR', 'PM', 'Av_VICM_MIN',
'Av_VICM_MAX']
minimize_metrics = ['Pw']
equate_metrics=[]
```

Define your initial design point , boundary and names

```
# ==========================================================================
#  Define Initial State, Bounds, and Names
# ==========================================================================
x_initial = [Wd, Ld, Wm, Lm, Wss, Lss, WB]
x_names = ["Wd", "Ld", "Wm", "Lm", "Wss", "Lss", "WB"]

b_w = (0.3e-6, 50e-6)  # For now use W_min_tech = 300 nm , W_max = 50 um we don't want bigger area
b_l = (0.3e-6, 10e-6)   ## For now use L_min_tech = 300 nm , L_max_tech = 10 um

bounds = [b_w, b_l, b_w, b_l, b_w, b_l, b_w]
```

Define your mathematical function

$$\vec{y} = f(\vec{x})$$

```
# ==========================================================================
# Create generic f(x)
# ==========================================================================
def f_eval_5t(x):
    return evaluate_5t_ota_ltspice(
        Wd=x[0], Ld=x[1], Wm=x[2], Lm=x[3],
        Wss=x[4], Lss=x[5], WB=x[6],
        Ib=IB, CL=CL,Vicm=Vicm,
        TGT_VICM_MIN=TGT_VICM_MIN, TGT_VICM_MAX=TGT_VICM_MAX,
        TGT_VOUT_MIN=TGT_VOUT_MIN, TGT_VOUT_MAX=TGT_VOUT_MAX,
        BASE_DIR=BASE_DIR)
```

Step 6 : Execution

Run the optimizer as follows:

```
ga = UniversalGeneticOptimizer(
    f_eval=f_eval_5t,
    x_initial=x_initial,
    bounds=bounds,
    y_spec=y_spec,
    maximize_keys=maximize_metrics,
    minimize_keys=minimize_metrics,
    equate_keys=equate_metrics,
    max_evals= 30,
    pop_size=10,
    x_names=x_names    # Maps the array indices to physical names for the
report
)

# 5. Execute and Report
x_opt, optimized_y = ga.optimize()
ga.print_report()
```

Step 7 : Report Results

Report the output specs in a table as follows :

Both In the Anti Gravity Chat and in a notepad/.txt file that you will create , report the output results in a tabular form like this :

```
=================================================================
 FINAL OPTIMIZATION REPORT
=================================================================
--- OPTIMAL DESIGN PARAMETERS (x_opt) ---
  x[0] = Wd_opt  :=
  x[1] = Ld_opt  :=
  x[2]= Wm_opt  :=
  x[3]=Lm_opt  :=
  x[4]=Wss_opt  :=
  x[5]=Lss_opt  :=
  x[6]=WB_opt:=
```

```
--- SPECIFICATION PERFORMANCE ---
  Spec            | Target       | Achieved     | Status
  -----------------------------------------------------------
  Av              |              |              |  [PASS/FAIL]
  UGF             |              |              |  [PASS/FAIL]
  CMRR            |              |              |  [PASS/FAIL]
  PM              |              |              |  [PASS/FAIL]
  Pw              |              |              |  [PASS/FAIL]
  Av_VICM_MIN     |              |              |  [PASS/FAIL]
  Av_VICM_MAX     |              |              |  [PASS/FAIL]
=================================================================
```

Step 8 : Verify Optimizer Result using Circuit Wrapper

Run this code in the same place where your math function is defined

```
y_opt=f_eval_5t(x_opt)
print("\n" + "=" * 65)
print(" Verify_Optimizer_Output")
print("=" * 65)
print(f"  1.  Av              : {y_opt['Av']:.4f} dB")
print(f"  2.  BW              : {y_opt['BW']/1e3:.2f} kHz")
print(f"  3.  UGF             : {y_opt['UGF']/1e6:.2f} MHz")
print(f"  4.  CMRR            : {y_opt['CMRR']:.4f} dB")
print(f"  5.  PM              : {y_opt['PM']:.2f} deg")
print(f"  6.  Power           : {y_opt['Pw']*1e6:.2f} uW")
print(f"  7.  Slew Rate       : {y_opt['Slew_Rate']/1e6:.2f} V/us")
print(f"  8.  Av(@TGT_VICM_MIN) : {y_opt['Av_VICM_MIN']:.2f} dB")
print(f"  9.  Av(@TGT_VICM_MAX) : {y_opt['Av_VICM_MAX']:.2f} dB")
print(f"  10. VOUT_MIN Err    : {y_opt['VOUT_MIN_Err']*1000:.2f} mV")
print(f"  11. VOUT_MAX Err    : {y_opt['VOUT_MAX_Err']*1000:.2f} mV")
print(f"  12. VOUT_CM         : {y_opt['VOUT_CM']:.2f} V")
print("=" * 65)
```

Observe whether specs are achieved or not and print the output in another .txt file call it
OptimizerVerify.txt and also print it in the Chat .

```
--- Verify Optimizer Output ---
  Spec              | Target        | Achieved      | Status
  -----------------------------------------------------------
  Av                |               |               |  [PASS/FAIL]
  UGF               |               |               |  [PASS/FAIL]
  CMRR              |               |               |  [PASS/FAIL]
  PM                |               |               |  [PASS/FAIL]
  Pw                |               |               |  [PASS/FAIL]
  Av_VICM_MIN       |               |               |  [PASS/FAIL]
  Av_VICM_MAX       |               |               |  [PASS/FAIL]
=================================================================
```